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1 of 3: ald rin
SAKE-IoT: Session Amortization for Post-Quantur

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4) **45.1% Per-Packet Bandwidth Reducti**
LDPC CT_0) → 224 bits (AES-GCM
5) **33.1× Clock Cycle Reduction:** ≈2.
LDPC HE) → ≈74K cycles (AEAD).
6) **Epoch-Bounded Forward Secrecy**
security property formally proven here

II. Related Work

A. Lattice-Based Authentication for IoT

Ring-LWE-based schemes offer the most
tational profile for IoT authentication. Li et
a lattice-based PAKE. Wang et al. [17] intro
Ring-LWE authentication. RLizard [18] dem
Ring-LWE key encapsulation but relied
polynomial multiplication. Importantly, **no**
amortize post-quantum costs over a sess
exchange requires a full handshake.

B. Code-Based Encryption for IoT

Hu et al. [19] analyzed QC-LDPC KEM
not target the IoT low-RAM profile. Kuma
closest prior work; we build directly upon th
extend the data phase with amortized AEAD

C. Session Management and Amortization

TLS 1.3 session resumption [4] motivate
targets classical cryptography on high-end ha
authentication schemes [21] use symmetric
derive them from classical (ECC) handshak
post-quantum foundation. To the best of our k
IoT is the **first scheme to amortize Ring-LW**
epoch establishment over a lightweight AES
session on constrained IoT hardware.

D. Differentiation from TLS 1.3 and DTLS

III. Preliminaries

A. Notation

B. Lattice and LWE Foundations

A *lattice* Λ is given by $\Lambda = \{a_1\beta_1 + \cdot$
$\mathbb{Z}\} \subset \mathbb{R}^n$, where $\beta_1, \ldots, \beta_k \in \mathbb{R}^n$ are line
vectors (basis) and n is the ambient dir
LWE [10]: given $(R_n, T_n = R_n\delta_n + \varepsilon_n)$, recov
LWE: distinguish LWE samples from uniform
Theorem 3 (EB-FS).

C. Ring-LWE

Ring-LWE is defined over $R_q = \mathbb{Z}_q[u]/$
$R_n \subset R^n$, and error $\varepsilon_n \leftarrow \mathcal{G}_\sigma^n$, pairs
od q) [10], [11]. The F
Theorem 3: Pr[findSc

1 Integrity Flag for Review

Suspicious text manipulations found. Review Flags as a priority.

Integrity Flag

Replaced characters

Letters are swapped with similar characters from another alphabet.

39 suspect characters on 7 pages

Common: 25

Greek: 14

What are replaced characters?

An attempt to hinder similarity detection by replacing characters with similar looking ones from different alphabets or character sets.

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